(i) Quartiles:

A measure which divides an array into four equal parts is known as quartiles.

Each portion contains equal number of items. The first, second and third points are termed as first quartile (Q_1), second quartile (Q_2) (better named as median) and the third quartile(Q_3). The first quartile (Q_1) or lower quartile, has 25% of the items of the distribution below it and 75% of the items are greater than it. Q_2(median) the second quartile or median has 50 % of the observations above it and 50% of the observations below it. The upper quartile or third quartile(Q_3) has 75% of the items of the distribution below it and 25% of the items are above it. Similarly the other two positional measures can be defined.

(ii) Deciles:

A measure which divides an array into ten equal parts is known as deciles.

That is Deciles are the values which divides the series into ten equal parts. We get nine dividing positions namely $D_1, D_2, \ldots, D_9$ which are called as deciles. Therefore there are nine deciles. It is to be noted that D_5 is equal to median.

(iii) Percentiles:

A measure which divides an array into hundred equal parts is known as Percentiles.

That is percentiles are the values which divides the series into hundred equal parts. We get 99 dividing positions $P_1, P_2, \ldots, P_{99}$ which are called as percentiles. Therefore there are 99 percentiles. It is to be noted that P_{50} is equal to median.

8.1.3 Computations for Related positional measure

The procedure for computing quartiles, deciles and percentiles are the same as the median.

(i) Ungrouped data:

Steps:

1. Arrange the data either in ascending or descending order of magnitude.

2. Apply the formula

Q_1 = Size of $\left(\frac{n+1}{4}\right)^{th}$ value

Q_3 = Size of $\left(\frac{3(n+1)}{4}\right)^{th}$ value

D_1 = Size of $\left(\frac{n+1}{10}\right)^{th}$ value

D_2 = Size of $2\left(\frac{n+1}{10}\right)^{th}$ value

P_{60} = Size of $60\left(\frac{n+1}{100}\right)^{th}$ value

P_{99} = Size of $99\left(\frac{n+1}{100}\right)^{th}$ value

DO YOU KNOW?

$Q_2 = D_5 = P_{50}$ = **Median**

Example 8.1

Find D_2 and D_6 for the following series 22, 4, 2, 12, 16, 6, 10, 18, 14, 20, 8

Solution:

Here n = 11 observations are arranged into ascending order

2, 4, 6, 8, 10, 12, 14, 16, 18, 20, 22

D_2 = size of $2\left(\frac{n+1}{10}\right)^{th}$ value

D_6 = size of $6\left(\frac{n+1}{10}\right)^{th}$ value

D_2 = size of 2.4th value $\approx$ size of 2nd value = 4

D_6 = size of 7.2^{th} value $\approx$ size of 7th value =14

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